Cloud Unlock · CloudAdhar and AWS Student Builder Group, MIT ADT University

Day 1: Cloud Unlock Kickoff - Careers, Certifications and a Safe AWS Account

Saturday, 10 October 2026, 7:00-9:00 PM IST · Trainer: Gangadhar Ure

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Live class: Saturday 10 Oct 2026, 7:00-9:00 PM IST
Exam focus: Domain 1 (Cloud Concepts), Domain 2 (Security), Domain 4 (Billing)
Exam tasks: 1.1 · 2.3 · 4.2 (plus 1.2 and 1.4 as reading)

Today covers: cloud careers and jobs · AWS certifications · opportunities · networking · learning and projects · live Q&A · and your first practical: a safe AWS account.

Pre-work: create your AWS account before the class (Lab 1 in 03-account-setup-console-tour.md) and keep an authenticator app ready on your phone.

Practice Sequence

Seq	Practice	File
01	Why cloud and the problems it solves	01-cloud-foundations.md
02	Careers, certifications, opportunities and networking	02-well-architected-and-careers.md
03	Create your account and tour the Console	03-account-setup-console-tour.md
04	Root MFA and an IAM admin user (budget and IAM group on Day 2)	04-account-security-lab.md

Live class: keep practical-steps.md open and tick each step with the trainer.

By The End Of Today

- You can explain cloud computing using one Amazon festival sale night: scalability, elasticity, high availability, reliability, security and pay-per-use.
- You know four cloud roles and the skills fresher jobs ask for.
- You know the AWS certification levels and the CLF-C02 exam format.
- You know two free places to learn AWS and two ways to meet the cloud community.
- Read after class: cloud economics, IaaS/PaaS/SaaS and the six pillars (taught again in S7, S10 and S11).
- You can sign in, switch Regions and find your Free Tier or credits page.
- Your root user has MFA, and you have an IAM admin user for daily work.
- Also shown today: Region, Availability Zone, edge location, Local Zone and Wavelength Zone (terms).
- You can name tasks only the root user can do, and tell Budgets, Cost Explorer and Pricing Calculator apart.

Today's Deliverables

- Screenshot of Console Home with the Region visible (account ID hidden)
- Screenshot showing root MFA assigned (no QR, no codes)
- Screenshot of the IAM user list (account ID hidden)
- Account security checklist ticked in `notes.md`
- Career path notes in `notes.md` (exam details, job skills, a user group near you) from 02-well-architected-and-careers.md
- Connect with 3 classmates on LinkedIn
- Day 1 LinkedIn post with `#CloudUnlock`: see 10-linkedin-post.md

Day 1 - Cloud Foundations

Goal: understand what the cloud is and which real problems it solves.

Exam tasks: 1.1 (benefits of the AWS Cloud), 1.4 (cloud economics), 3.1 (deployment models)

Before You Start

Write answers in your own words before reading:

1. What is cloud computing? _____
2. Why would a company use cloud instead of its own servers? _____
3. Name one service you already use that runs on the cloud. _____

The Story - Sale Night On A Shopping App

Think of **Flipkart Big Billion Days**, the **Amazon Great Indian Festival** or a **Myntra sale**. When the sale opens, people all over India open the app at the same moment. They search, add to cart and pay with **PhonePe** or **Google Pay**. A week later the rush is gone.

Now imagine **you** run an app like that. These are teaching numbers, not real company figures:

When	Shoppers on the app	Servers you need
Normal day	A few thousand	2
Sale opens	Lakhs at once	20
A week later	A few thousand again	2

With your own servers: you buy 20 servers for the sale, wait weeks for them to arrive, and pay for all 20 all year. For 50 weeks, 18 of them sit idle.

With the cloud: you add servers when the sale opens, remove them when it ends, and pay only for the hours you used them.

Walk through one night - Amazon Great Indian Festival

You have been waiting for a phone deal. Follow your own steps and see what the app has to do at each one:

Your step	What the app must do	Cloud word
11:59 PM - you and lakhs of others open the app and keep refreshing	Handle a sudden rush without slowing down	Scalability, elasticity
12:00 AM - the deal opens, you tap **Add to Cart**	Keep your cart even if one server fails	High availability
You pay with **PhonePe** or **Google Pay**	Never lose the order or the payment record	Reliability

Your step	What the app must do	Cloud word
You open **My Orders**	Show your orders and address only to you	Security
A week later the sale ends	Remove the extra servers and stop paying for them	Cost (pay for what you use)

The same thing happens in **Flipkart Big Billion Days** and the **Myntra** sale.

These five steps are the five problems the cloud solves (next section). We describe only what you see as a user, not how these companies build their systems.

Five Problems The Cloud Solves

Problem	Without cloud	With AWS	Service (later)
Scaling (autoscaling)	Buy for the peak	Capacity grows and shrinks with traffic	EC2 Auto Scaling, Elastic Load Balancing
Cost	Large upfront spend	Pay for what you use	Budgets, Cost Explorer
Availability	One data centre fails, site down	Spread across Availability Zones	Multi-AZ, Route 53, CloudFront
Security	You secure everything alone	Shared responsibility	IAM, KMS, Security Groups
Reliability	Manual backups	Automated backups and replication	S3, RDS, AWS Backup

Six Advantages Of Cloud Computing

1. Trade fixed expense for variable expense.
2. Benefit from massive economies of scale.
3. Stop guessing capacity.
4. Increase speed and agility.
5. Stop spending money running and maintaining data centres.
6. Go global in minutes.

Four Words The Exam Loves

Word	Meaning	Real-app example
High availability	The app keeps running when one part fails	The Amazon app still opens on sale night if one data centre has a problem
Elasticity	Resources grow **and shrink** automatically with demand	2 servers normally, 20 when the sale opens, back to 2 after
Scalability	The system can handle more load by adding resources	Ready for more shoppers as the app grows every year

Word	Meaning	Real-app example
Agility	Try new things fast and cheaply	Try a new "Buy Now" button for one week, remove it if nobody uses it

Global reach is a benefit too: deploy close to users anywhere in minutes.

Cloud Economics

Idea	On-premises	AWS Cloud
Fixed vs variable cost	Big upfront spend (CapEx) on servers, buildings, power, cooling, staff	Pay for what you use (OpEx)
Hidden on-prem costs	Hardware refresh, electricity, cooling, floor space, security guards, data centre staff	Included in the service price
Economies of scale	Your small volume, your price	AWS buys at huge volume and passes savings on
Rightsizing	Hard: you already bought the server	Pick the right size, change it any time
Automation	Manual setup, slow, error-prone	Scripts and IaC: faster, repeatable, fewer mistakes
Licensing	You own licences	**BYOL** (bring your own licence) or **licence included** in the price

Exam pointer:

- "Reduce upfront cost" or "trade capital expense for variable expense" = cloud economics.
- "Resources match demand automatically" = elasticity.
- "Already own Windows or Oracle licences" = BYOL.
- "Instance is too big for the workload" = rightsizing.

Service Models

- **IaaS**: you manage OS and apps (Amazon EC2).
- **PaaS**: you manage app and data; the platform is managed (Elastic Beanstalk, RDS).
- **SaaS**: you just use it (Gmail, Microsoft 365).

Deployment Models

- Cloud, on-premises (private cloud), and hybrid (both together).

Exam pointer:

- Pay-as-you-go and no upfront cost point to cloud benefits.
- Own data centre plus AWS together means hybrid.
- You manage the OS means IaaS.

Practice - Explore The Official AWS Pages

Practice in this order:

1. Open What is cloud computing?. Find the section on benefits.
2. Open Types of cloud computing. Find IaaS, PaaS and SaaS.
3. Open AWS customer stories. Filter by one industry you like (for example Education, Media or Financial Services).
4. Open one story. Read the headline and the results.

Add to your `notes.md`:

```
Cloud in one line (my words):
One benefit I found on the AWS page:
Customer story I read (company + one result it mentions):
```

Expected result: three short lines in your notes, written in your own words.

Official Links

Topic	**Link**
What is cloud computing	https://aws.amazon.com/what-is-cloud-computing/
Types of cloud computing (IaaS, PaaS, SaaS)	https://aws.amazon.com/types-of-cloud-computing/
Six advantages of cloud computing	https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html
AWS customer stories	https://aws.amazon.com/solutions/case-studies/
AWS Pricing overview	https://aws.amazon.com/pricing/

Day 1 - Careers, Certifications, Opportunities and Networking

Goal: know where cloud skills can take you, which certification to start with, where to learn for free and how to meet people in the cloud community.

Certification Path

- Foundational: **Cloud Practitioner (CLF-C02)** (our goal), AI Practitioner.
- Associate: Solutions Architect, Developer, SysOps/CloudOps Engineer, Data Engineer.
- Professional and Specialty: senior roles and specialisation.
  Check aws.amazon.com/certification for the current list, prices and renewal rules.

The CLF-C02 Exam At A Glance

Item	Detail
Questions	65 (50 scored + 15 unscored; you cannot tell which)
Time	90 minutes
Question types	Multiple choice (one answer) and multiple response (two or more answers)
Pass mark	700 out of 1000 (scaled score); you only need to pass overall
Guessing	No penalty, so never leave a question blank
Domains	Cloud Concepts 24% · Security and Compliance 30% · Cloud Technology and Services 34% · Billing, Pricing and Support 12%

Cloud Careers And Jobs

Role	What you do	Common starting skills
Cloud Support Associate	Help customers fix cloud problems	AWS basics, Linux, networking
Cloud Engineer	Build and run systems on AWS	AWS, Linux, scripting
DevOps Engineer	Automate how apps are built and released	Git, CI/CD, containers, IaC
Solutions Architect	Design the whole system	Broad AWS knowledge, communication

Data Engineer and Security Engineer are other common paths.

Employers look for hands-on skills plus a certification. That is why every week has a practical and a GitHub submission: it becomes your portfolio.

Opportunities For Students

Opportunity	What it is	Link
AWS Skill Builder	Free AWS courses and practice questions	https://skillbuilder.aws/
AWS Educate	Free cloud learning for students	https://aws.amazon.com/education/awseducate/
AWS Builder Center	Read and share articles by builders	https://builder.aws.com/
AWS Community Builders	A programme for people who build and share what they learn	https://aws.amazon.com/developer/community/community-builders/
Your own projects	A website for your college club or event, built on AWS	Your GitHub

Read each programme's current rules on its official page.

Networking

- **LinkedIn:** post what you learn every week, in your own words, with `#CloudUnlock`. Connect with classmates.
- **AWS communities:** AWS User Groups, AWS Community Day, AWS Summits. Go to one event this year and talk to three people.
- **This group:** the Cloud Unlock WhatsApp group, Meetup and YouTube live chat.
- **Ask good questions:** AWS re:Post, the official AWS Q&A community.

Lab - Explore Your Career Path

Practice in this order:

1. Open AWS Certification. Find the four levels: Foundational, Associate, Professional, Specialty.
2. Open AWS Certified Cloud Practitioner. Note the exam code, time and number of questions.
3. On the same page, open the **exam guide**. Find the four domains and their weights.
4. Open AWS Skill Builder. Sign up for free. Search `Cloud Practitioner` and find one free course.
5. Open LinkedIn Jobs or Naukri. Search `AWS fresher`, location `India`.
6. Open two job posts. Write down the skills they ask for.
7. Open AWS Educate and AWS User Groups. Find one user group near you.
8. On LinkedIn, connect with 3 classmates from this series.

Add to your `notes.md`:

```
Exam code, time, number of questions:
Four exam domains:
Free Skill Builder course I found:
Job 1 skills:
Job 2 skills:
One skill I will build in this series:
AWS user group near me:
3 classmates I connected with (first names only):
```

Expected result: your own small career plan in `notes.md`. Do not post salary or personal details.

If you get stuck, write what you completed:

```
I found the exam details. I could not sign up for Skill Builder yet.
```

Reflection (add to your notes)

Which role interests you most and why? Which community event will you try to attend?

Read After Class - The Six Pillars Of The AWS Well-Architected Framework

We teach this in detail in S10. Read it now so the names are familiar (exam task 1.2).

Pillar	Question it asks	Benefit	Example
Operational Excellence	Can we run, monitor and improve it?	Fewer outages	Automate deployments, monitor
Security	Is data and access protected?	Trust, compliance	Least-privilege IAM, encryption
Reliability	Does it recover and keep working?	Less downtime	Multi-AZ, backups
Performance Efficiency	Do we use the right resources?	Fast, no waste	Right instance type, caching
Cost Optimization	Do we pay only for value?	Lower bills	Right-size, turn off idle
Sustainability	Do we minimise environmental impact?	Lower energy and cost	Efficient resources, scale down

Exam pointer: "recover from failure" = Reliability. "Reduce spend" = Cost Optimization. "Protect data" = Security.

Common confusion:

- Performance Efficiency is about using the **right** resources well (speed). Cost Optimization is about **not paying** for waste.

- Operational Excellence is about **running and improving** the workload. Reliability is about **recovering** from failure.
- Sustainability is about **energy and environmental impact**, often by removing idle resources.

Official Links

Topic	Link
Six pillars of the Well-Architected Framework	https://docs.aws.amazon.com/wellarchitected/latest/framework/the-pillars-of-the-framework.html
AWS Well-Architected overview	https://aws.amazon.com/architecture/well-architected/
AWS Educate	https://aws.amazon.com/education/awseducate/
AWS Community Builders	https://aws.amazon.com/developer/community/community-builders/
AWS User Groups	https://aws.amazon.com/developer/community/usergroups/
AWS Community Day	https://aws.amazon.com/events/community-day/
AWS re:Post	https://repost.aws/
AWS Certification (all levels)	https://aws.amazon.com/certification/
AWS Certified Cloud Practitioner (exam page and guide)	https://aws.amazon.com/certification/certified-cloud-practitioner/
AWS Skill Builder (free training)	https://skillbuilder.aws/
AWS Training and Certification	https://aws.amazon.com/training/

Day 1 - Account Setup and Console Tour

Exam tasks: 3.1 (ways to access AWS), 4.3 (AWS Support plans)

Practice Rule

Use your **own** AWS account. Never share your email, password, OTP, card or account ID.

Practice in this order:

1. Create your AWS account (before the live class).
2. Sign in as the root user.
3. Tour the Console.
4. Take one safe screenshot.

Lab 1 - Create Your AWS Account

Please do this **before the Day 1 live class**. Verification and activation can take time. Skip if you already have an account.

1. Open aws.amazon.com and choose **Create an AWS Account**.
2. Enter your personal email and an account name, for example `cloudadhar-learning`.
3. Open your email, copy the verification code and paste it.
4. Create a strong, unique root password. Save it in a password manager.
5. Choose the **Free plan** for learning. It gives credits for a limited time. Check the Billing page for current terms.
6. Choose **Personal**. Enter your contact details. Accept the agreement.
7. Add a payment method. AWS uses it to verify identity. Never share these details.
8. Verify your phone by SMS or voice call.
9. Choose the **Basic support** plan. It is free.
10. Wait for activation. It takes minutes, sometimes up to 24 hours.
11. Go to the AWS Console. Choose **Root user** and sign in.

Expected result: you see **Console Home**.

Screens and plan names can change. Follow the on-screen text.

Lab 2 - Console Tour

Test:

1. Look at the top right. Find the **Region** name.
2. Click it and choose **Asia Pacific (Mumbai) ap-south-1**.
3. Click the search bar at the top. Type `EC2` and open it. Click the AWS logo to go back.
4. Search `S3` and open it. Go back.

5. Switch the Region to **US East (N. Virginia) us-east-1**. Notice the Region name change. Switch back to **Mumbai**.
6. Search `Billing` and open **Billing and Cost Management**.
7. Find **Free Tier** or **Credits** in the left menu. Look at what is used and what is left.
8. Click **Services** (or the grid icon). Find one service in each group: Compute, Storage, Database, Networking, Security.

Expected result: you can say which Region you are in and where to find your credits.

Add to your `notes.md`:

```
My Region:
One service each - Compute: / Storage: / Database: / Networking: /
Security:
What surprised me in the Console:
```

Deliverables

- Screenshot of Console Home with the Region visible (account ID hidden)
- One sentence: what surprised you in the Console?

If You Get Stuck

Problem	Try this
No OTP or SMS	Wait 2 minutes, retry, or choose voice call
Card verification fails	Try another card or bank. Never post card details in chat
Account "pending activation"	Normal. Wait up to 24 hours, then sign in again
Console in another language	Gear icon (Settings) → Language

Submit what you completed with a note:

```
My account is still pending activation. I watched the Console tour and will
finish tomorrow.
```

Next: secure your account right away in 04-account-security-lab.md.

Official Links

Topic	Link
Create and activate an AWS account	https://repost.aws/knowledge-center/create-and-activate-aws-account
Getting started with your AWS account	https://docs.aws.amazon.com/accounts/latest/reference/getting-started.html
AWS Free Tier	https://aws.amazon.com/free/

Topic	Link
AWS Free Tier plans and credits	https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/free-tier-plans.html
AWS Management Console guide	https://docs.aws.amazon.com/awsconsolehelpdocs/latest/gsg/what-is.html
AWS Support plans	https://aws.amazon.com/premiumsupport/plans/

Day 1 - Account Security Lab

Goal: create a safe AWS account foundation for the next five weeks.

Exam tasks: 2.3 (root user protection, MFA, IAM users and groups), 4.2 (AWS Budgets)

In class: Labs 1 and 3 on **Day 1** (Saturday). Labs 2 and 4 at the start of **Day 2** (Sunday).

Why The Root User Is Special

The root user is the email you signed up with. It has full access to everything and cannot be limited by IAM policies.

Some tasks only the root user can do, for example:

- Change account settings: account name, root email, root password
- Close the AWS account
- Change or cancel the AWS Support plan
- Restore permissions when the only IAM admin is locked out
- Turn on IAM user access to the Billing console

Everything else should be done with an IAM user (Lab 3 below) or an IAM role (Week 2).

Exam pointer: "protect the root user" = turn on MFA, do not create root access keys, do not use root for daily work, use a strong unique password.

Practice Rule

Sign in as the root user for Labs 1-3. Do not create any servers or storage.

Practice in this order:

1. Install an authenticator app on your phone.
2. Turn on MFA for the root user. Test it by signing in again. (Day 1)
3. Check that there are no root access keys. (Day 1)
4. Create an IAM admin user for daily work. (Day 1)
5. Create a budget alert. (Day 2)
6. Move admin access to a group, add MFA to the IAM user, and sign in as the IAM user. (Day 2)

Before You Start

You need:

- Your root email and password
- A phone with an authenticator app (Google Authenticator, Microsoft Authenticator or similar)
- Access to your email

Lab 1 - Secure The Root User

Create:

- MFA device name: `root-mfa-phone`

Steps:

1. Sign in to the AWS Console as the **Root user**.
2. Click your account name at the top right. Choose **Security credentials**.
3. Find the **Multi-factor authentication (MFA)** section. Click **Assign MFA device**.
4. Device name: `root-mfa-phone`.
5. Choose **Authenticator app**. Click **Next**.
6. Click **Show QR code**. Open your authenticator app and scan it. Do **not** screenshot the QR.
7. Type the 6-digit code from the app into **MFA code 1**.
8. Wait for the code to change (about 30 seconds). Type the new code into **MFA code 2**.
9. Click **Add MFA**.

Test:

1. Click your account name → **Sign out**.
2. Sign in again as the root user.
3. AWS asks for an MFA code. Type the code from your app.

Expected result: you are signed in, and the MFA device `root-mfa-phone` is listed under **Security credentials**.

Check access keys:

1. On the same **Security credentials** page, find **Access keys**.
2. It should say there are no access keys. Do not create any.
3. If you find an old key you do not use, delete it. If you are unsure, ask in the community first.

Stop using root for daily work. We create a safer IAM user in Week 2.

Deliverables:

- Screenshot showing root MFA assigned (no QR, no codes, account ID hidden)
- Short note: why root should not be used daily

Lab 2 - Budget Alert (Day 2, first thing in class)

Create:

- Budget name: `cloud-unlock-learning-budget`
- Amount: USD 5 (or the Zero spend template)

Steps:

1. Search `Budgets` in the Console. Open **Billing and Cost Management → Budgets**.
2. Click **Create budget**.
3. Choose **Use a template (simplified)**.
4. Choose **Zero spend budget**, or **Monthly cost budget** with amount `5`.
5. Budget name: `cloud-unlock-learning-budget`.
6. Email recipients: your email.
7. Click **Create budget**.

Test:

1. Go back to **Budgets**.
2. Find `cloud-unlock-learning-budget` in the list.

Expected result: the budget appears in the list with your amount.

Remember: budgets **alert** you. They do **not** stop spending. Alert emails can arrive late, so always clean up after labs.

Deliverables:

- Screenshot of the budget list
- Short note: why monitor cost from day one (budgets alert; they do not stop spending)

Lab 3 - Create An IAM Admin User (Day 1)

Why: root is the master key. For daily work you use a separate **IAM user**, like a staff ID card instead of the owner's key.

Create:

- User name: `Cloud-Unlock` (or the name you used in class)
- Permission: `AdministratorAccess`

Steps (signed in as root):

1. Search `IAM` and open it. Left menu: **Users → Create user**.
2. User name: `Cloud-Unlock`.
3. Tick **Provide user access to the AWS Management Console**. If asked, choose **I want to create an IAM user**.
4. Console password: **Custom password**. Use a strong password you do not use anywhere else.
5. Click **Next**.
6. Permissions options: **Attach policies directly**. Search `AdministratorAccess` and tick it.
7. Click **Next → Create user**.
8. The last page shows the **Console sign-in URL**. Save it in your notes. It contains your **account ID**, so never share it or show it in a screenshot.

Expected result: `Cloud-Unlock` appears in **IAM → Users** with `AdministratorAccess`.

Deliverable: screenshot of the IAM user list (account ID hidden).

Lab 4 - Admin Access Through A Group (Day 2)

Better approach: attach policies to **groups**, then add users to groups. 100 engineers? One group, one set of permissions. Exam pointer: "manage permissions for many users" = **IAM groups**.

Practice in this order:

1. Create group.
2. Attach policy to group.
3. Add user to group.
4. Remove the policy attached directly to the user.
5. Add MFA to the user.
6. Sign in as the user and test.

Create:

- Group: `AdminGroup`
- Policy: `AdministratorAccess`
- User: `Cloud-Unlock` (from Lab 3)
- MFA device: `iam-admin-mfa-phone`

Steps (signed in as root):

1. **IAM → User groups → Create group**. Name: `AdminGroup`.
2. Add users: tick `Cloud-Unlock`.
3. Attach permissions policies: tick `AdministratorAccess` → **Create user group**.
4. **IAM → Users → `Cloud-Unlock` → Permissions**: select `AdministratorAccess` **Attached directly → Remove**. The user keeps access through the group.
5. Same user → **Security credentials → Assign MFA device** → `iam-admin-mfa-phone` → **Authenticator app** → scan → two codes → **Add MFA**.
6. Optional, recommended: **IAM → Dashboard → AWS Account → Account alias → Create** (for example `cloudunlock-yourname`). The sign-in URL then shows the alias instead of the account ID.

Test:

- Sign out of root. Open the IAM sign-in URL and sign in as `Cloud-Unlock` with password and MFA code.
- Confirm the top right shows `Cloud-Unlock`, not root.
- Confirm **IAM → Users → `Cloud-Unlock` → Permissions** shows `AdministratorAccess` attached via `AdminGroup`.

From now on, do all practice as `Cloud-Unlock`. Root stays locked with MFA.

Add these in your submission:

- Screenshot of `AdminGroup` with `Cloud-Unlock` inside (account ID hidden).
- Short note: why permissions go on groups, not users.

Budgets Or Cost Explorer?

Tool	Use it to	Think of it as
AWS Budgets	Get an alert **before or when** spend crosses a limit	Smoke alarm
AWS Cost Explorer	See and analyse **past** spend and forecasts	Bank statement
AWS Pricing Calculator	Estimate cost **before** you build	Shopping quote

Account Security Checklist

Tick these in your `notes.md` (do not share any IDs):
□ Root user has MFA
□ No root access keys exist
□ Root password is strong and stored safely (for example in a password manager)
□ IAM admin user `Cloud-Unlock` created (Day 1)
□ Budget alert created with my email (Day 2)
□ Admin access moved to group `AdminGroup`, MFA on the IAM user, signed in as the IAM user (Day 2)
□ I know which Region I work in (Mumbai `ap-south-1`)
□ I know where to find Billing and Free Tier or credits

If You Get Stuck

Problem	Try this
"Codes not valid"	Set your phone time to automatic. Wait for a new code before typing the second one
Lost the phone with MFA later	Use **Troubleshoot MFA** on the sign-in page (email and phone check)
Budgets page shows an error	Make sure you signed in as the root user
IAM user cannot see Billing or Budgets	Normal by default. Do the budget as root, or (root only) Account → turn on **IAM user and role access to Billing information**
Forgot the IAM sign-in URL	Sign in as root → **IAM** → **Dashboard** shows the sign-in URL
Alias name already taken	Account aliases are unique worldwide. Add your name or a number

Add a screenshot of where you got stuck, write 2-3 lines, and ask in the community. Do not skip the whole week.

> I turned on MFA. I got stuck creating the budget because the page showed an error.

Safety Note

Do not share root email, account ID, access keys, MFA QR code, OTP, payment details or detailed billing.

Official Links

Topic	Link
Root user best practices	https://docs.aws.amazon.com/IAM/latest/UserGuide/root-user-best-practices.html
Tasks that need the root user	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_root-user.html
Turn on MFA for the root user	https://docs.aws.amazon.com/IAM/latest/UserGuide/enable-virt-mfa-for-root.html
MFA in AWS	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa.html
Create an IAM user	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_users_create.html
Create IAM groups	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_groups_create.html
Add or remove users in a group	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_groups_manage_add-remove-users.html
MFA for an IAM user	https://docs.aws.amazon.com/IAM/latest/UserGuide/enable-virt-mfa-for-iam-user.html
Account alias	https://docs.aws.amazon.com/IAM/latest/UserGuide/console-account-alias.html
IAM best practices	https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html
AWS Budgets	https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html
AWS Cost Explorer	https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html
AWS Pricing Calculator	https://calculator.aws/